EE-170LComputer Programming Lab
Laboratory Session 07
One-Dimensional Array in C++

1 Objective

The students will learn and apply the

- What is Array?
- Declaration of Array
- Initialization of Array
- Accessing Elements of Array
- Copying Arrays
- Character Array (String)

2 What is an Array?

Array is a data structure in C++, which stores a fixed size **sequential collection** of elements of the **same type**. An array is used to store a collection of data. It is often useful to think of an array as a collection of variables of the same type. All arrays consist of contiguous memory locations. The lowest address corresponds to the first element and the highest address to the last element.

3 Declaration of Array

To declare an array in C++, the programmer specifies the **type** of the elements and the **number of elements** required by an array as follows:

Syntax:

```
dataType arrayName[arraySize];
```

This is called a single-dimension array. The arraySize must be an integer constant greater than zero and type can be any valid C++ data type. For example, to declare a 4-element array called age of type int, use this statement:

Example:

```
int age[4];
```

- **int :** Data type of array
- **age :** Name of array
- **int :** Size of array

4 Initialization of Array

It is like a variable, an array can be initialized. To initialize an array, we provide initializing values which are enclosed **within curly braces** in the declaration and placed following an equals sign after the array name. Here is an example of initializing an integer array.

Example:

```
int age[4]={23,34,64,75};
```

Example:

```
1  #include <iostream>
2  using namespace std;
3  int main()
4  {
5      // declaration of array
6      int age[4];
7      // initialization f array elements
8      age[0] = 23;
9      age[1] = 34;
10     age[2] = 64;
11     age[3] = 75;
12     return 0;
13 }
```

LabCodes/lab08p01.cpp

On line (6), an array of 4 integers is created. Values are assigned to each variable in the array on line (8) through line (11).

5 Accessing Elements of Array

In any point of a program in which an array is visible, we can access the value of any of its elements individually as if it was a **normal variable**, thus being able to both read and modify its value.

Syntax:

arrayName[index];;

Following the previous examples in which"age"had 4 elements and each of those elements was of type int, the name which we can use to refer to each element is the following:

Examples:

age[0]; // 1st element of array
age[1]; // 2nd element of array
age[2]; // 3rd element of array
age[3]; // 4th element of array

Note that array **index start with 0** in C++. For example, to store the value 75 in the third element of age , we could write the following statement:

Example:

age[2]=75;

And, for example, to store the value of the third element of age to a variable called a, we could write:

Example:

a = age[2];

6 Copying Arrays

Suppose that after filling our 4 element array with values, we need to copy that array to another array of 4 integers ? Try this:

Example:

```
#include <iostream>
using namespace std;
int main()
{
    // declaration of array
    int age[4];       // declare first array
    int same_age[4]; // declare second array
    int i = 0;
    // initialization f array elements
    age[0] = 23;
    age[1] = 34;
    age[2] = 64;
    age[3] = 75;

    for (; i < 4; i++)
    {
        same_age[i] = age[i];
    }

    for (i = 0; i < 4; i++)
    {
        cout << same_age[i] << "\t";
    }
        return 0;
    }
```

LabCodes/lab08p02.cpp

In the above program two arrays are created: age and same age. Each element of the age array is assigned a value. Then, in order to **copy the four elements** in age into the same age array, we must **do it element by element**. We have used for loop to access every element of array note that for loop takes value from 0 to 3.

Note: Like printing arrays, there is no single statement in the language that says "copy an entire array into another array". The array elements must be copied individually. Thus If we want to perform any action on an array, we must repeatedly perform that action on each element in the array.

Lab07:Lab Tasks

1. Array Declaration and Initialization

   - Declare an integer array called ”numbers” with a size of 5.
   - Initialize the array with values 10, 20, 30, 40, and 50.
   - Display the values of the array elements using a loop.

2. Accessing Array Elements

   - Declare a character array called ”message” to store a word of your choice.
   - Use a loop to access each character of the array and display it on a separate line.

3. Copying Arrays

   - Declare two integer arrays: ”source” and ”destination”, each with a size of 5.
   - Initialize the ”source” array with some values.
   - Use a loop to copy the elements of the ”source” array into the ”destination” array.
   - Display the values of both arrays to verify the copy operation.

4. Array Sum

   - Write a C++ program that prompts the user to enter 5 numbers.
   - Store the numbers in an integer array.
   - Calculate the sum of all the numbers in the array.
   - Display the calculated sum.